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Final Project- Step 6

Url:http://flip1.engr.oregonstate.edu:6367/

Executive Summary

Our website connects the Gogurt database to a client side web server that the company uses to make business decisions. We have made many changes to the database design based on peer feedback.

- The original database schema had some incorrect naming conventions such as some attributes being camel cases and others being underscored. We also had to fix the Customers table because it was not working due to a foreign key constraint.

- We added CASCADE statements to foreign keys so that the associated attributes in the other tables would be updated to NULL when the fk is deleted. For example, if a customer is deleted, the associated customerID in the sales table is set to NULL.

- We updated the UI to be more user friendly including spacing between all tables and forms and an easy to use nav bar on each page. The Gogurt page acts as an index page because it is the focal point of the website.

- Added optional relationships between Customers/Sales and Suppliers/Orders to allow for a nullable relationship. We also changed our UI to have the information needed for the SalesDetails/OrdersDetails pages be calculated by the backend rather than entered by the user.
- The ERD was updated to only include the primary keys of each entity. The Schema and ERD were both updated so relationship lines did not cross to improve readability.

- Intersection tables were added to Project Outline and to the Schema. These were left out before we had a full understanding of what intersection tables were and how they were described.

★ Project Outline

A) Overview

Gogurt Inc. has annual yogurt sales of $2 million. Gogurt Inc also has to negotiate with many different suppliers, orders and customers. A database driven website for Gogurt will record the *sales* of *Gogurt* to *customers.* It will also record the invoice from *Supplier* to *Gogurt* from *orders.* The database will provide separate entities for *Supplier, Orders, Customers,* and *Gogurt* (product). The database creates a primary key for each entity. Insertion of foreign and private keys is another major advantage of database driven back-ends. For example if we need to update the address of a customer in *customers.* The information will also be updated in *sales* because the primary key of customers is included as a foreign key in sales. If we have 500 customers who account for 10,000 total sales, the database will auto-index an, easy to read, ID for each entry. If a single customer has made 10 purchases, the database can manage this data easily. Business or other important decisions can be made regarding this information management. For instance, we can see who the most valuable customers are based on the amount of revenue they have attributed or possibly compare the price of a product from two different suppliers and order more product from the cheaper option.

B) Outline

Entities:

Gogurts: stores details of which Gogurt products the store is buying/selling

Attributes and Relationships:

- idGogurt: int, not NULL, PK
- flavor: VARCHAR(50)
- qtyOnHand: int
- price: float, not NULL
- Relationship: a M:M between Gogurt and Orders with idGogurt and idOrders being foreign keys to the intersection table, OrdersDetails. When ordering more Gogurt from a supplier, there can be orders with multiple Gogurt products and multiple Gogurt products on order.
- Relationship: a M:M between Gogurt and Sales with idGogurt and idSales being foreign keys to the intersection table, SalesDetails. When selling Gogurt to customers, there can be a sale with multiple Gogurt products and a Gogurt product in multiple sales.

Suppliers: stores details about Gogurt suppliers

Attributes and Relationships:

- ❖ idSupplier: int, not NULL, PK
- ❖ supplierName: VARCHAR (255), not NULL
- ❖ email: VARCHAR(100), not NULL
- ❖ Relationship: a 1:M relationship with the Orders entity. There must be one supplier per order, but a supplier can fulfill many orders. Use idSupplier as foreign key in the Orders table.

Customers: stores details about Gogurt customers.

Attributes and Relationships:

- ❖ idCustomer: int, not NULL, PK
- ❖ customerName: VARCHAR(100), NULL
- ❖ email: VARCHAR(100), NULL
- ❖ address: VARCHAR(100)
- ❖ Relationship: has a 1:M relationship with the Sales entity. There must be one customer per Sales entry, but there does not have to be a Sales entry to add a customer. Use idCustomer as foreign key in the Sales table.

Orders: stores details about past/current orders between Gogurt and the suppliers.

Attributes and Relationships:

- ❖ idOrder: int, not NULL, PK
- ❖ idSupplier: int, FK (optional)
- ❖ orderDate: date, not NULL
- ❖ Relationship: has 1:M relationship with supplier, idSuppliers is a foreign key to orders. There must be only one supplier per Order entry
- ❖ Relationship: has M:M with the Gogurt entity where an intersection table, OrdersDetails, contains idOrder (Orders PK) and idGogurt (Gogurts PK) as foreign keys. Each order must have one or more kinds of Gogurt products and each Gogurt product can have multiple orders.

Sales: stores details of store Sales to customers

Attributes and Relationships:

- ❖ idSale: int, not NULL, PK
- ❖ idCustomer: int, FK (optional)
- ❖ shipDate: date
- ❖ subTotal: float

- Relationship: has 1:M relationship with Customer, idCustomer is a foreign key to Sales. A customer can participate in multiple sales, but each sale may only have 1 customer.
- Relationship: has M:M relationship with the Gogurt entity where an intersection table, SalesDetails, contains idSale (Sales PK) and idGogurt (Gogurts PK) as foreign keys. Each sale must have one or more kinds of Gogurt products and each Gogurt product can have multiple sales.

SalesDetails: intersection table for M:M relationship between Sales and Gogurt

Attributes and Relationships:

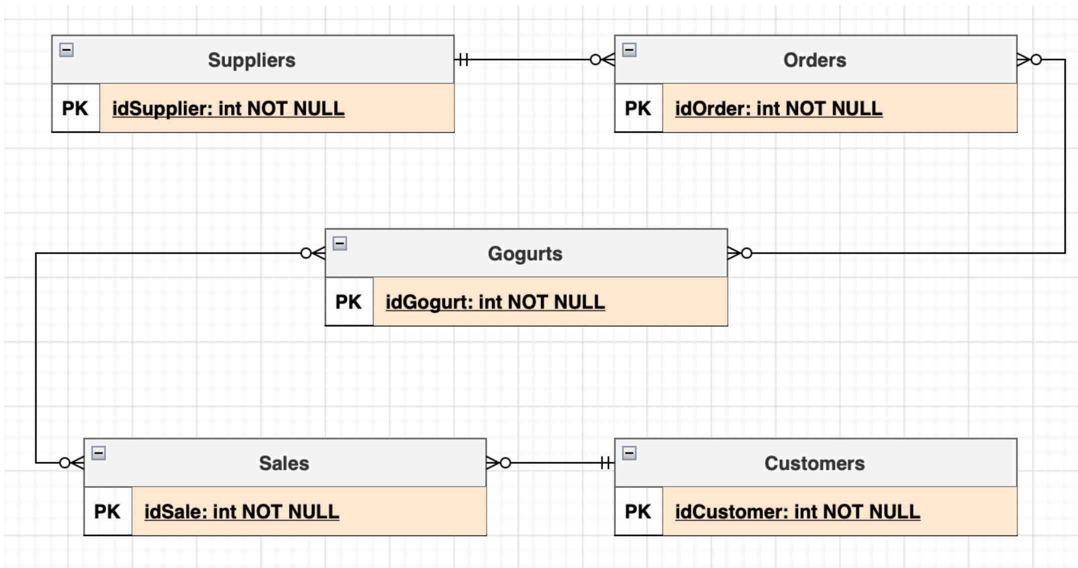
- idSalesDetail: int, not NULL
- sid: int, not NULL, FK (sale id)
- gid: int, not NULL, FK (gogurt id)
- salesDate: date, not NULL
- orderQty: int
- unitPrice: float
- lineTotal: float
- Relationship: has 1:M relationship with the Gogurt entity and a 1:M relationship with the Sales entity. Keeps track of Gogurt sales and Gogurt products that have been sold.

OrderDetails: intersection table for M:M relationship between Orders and Gogurt

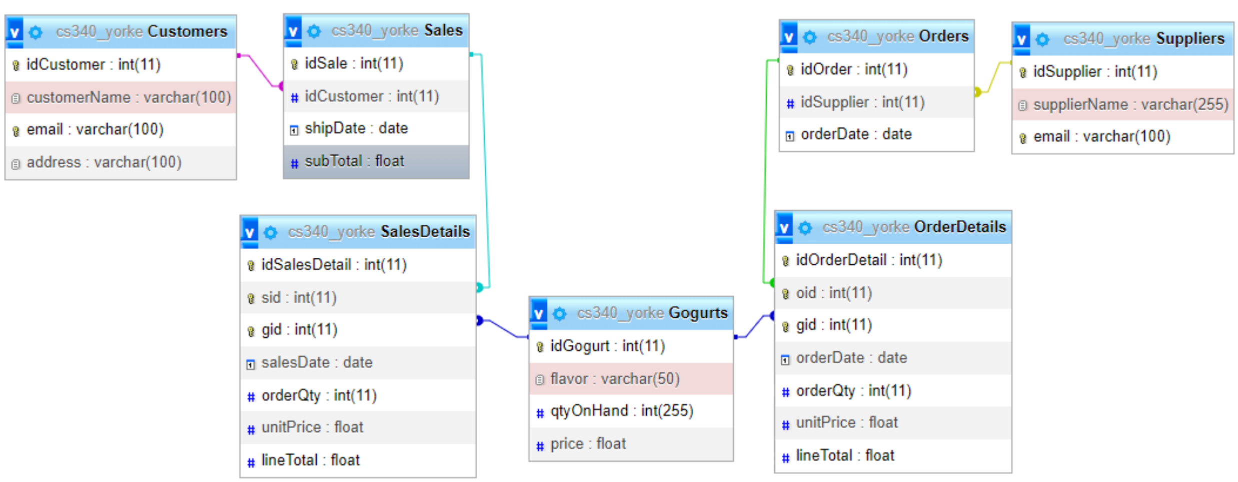
Attributes and Relationships:

- inOrderDetail: int, not NULL, PK
- oid: int, not NULL, FK (order id)
- gid: int not NULL, FK (gogurt id)
- orderDate: date
- orderQty: int
- unitPrice: float
- lineTotal: float
- Relationship: has 1:M relationship with the Gogurt entity and a 1:M relationship with the Orders entity. Keeps track of Gogurt orders and Gogurt products that have been ordered.

C) Entity-Relationship Diagram



D) Schema



E) Sample Data

Gogurts

Flavor	qtyOnHand	Price
Cherry	25	2.99
Watermelon	150	2.99
Green Apple	99	3.49

Customers

customerName	email	address
Jerry West	jwest@sample.com	123 Nice Ave
Michael East	meast@sample.com	1212 Wth Pl
Sara South	ssouth@sample.com	633 Yolo Dr

Suppliers

supplierName	email
Good Yogurt Co.	goodyogurt@sample.com
All Curds Inc.	allcurds@sample.com
South Sides Products	sideproducts@sample.com

Orders

productNum	idSupplier
10	3
3	1
6	2

Sales

idCustomer	shipDate	subTotal
1	2022-02-20	100.25
1	2022-12-10	235.98
2	2022-08-21	744.35
3	2023-01-06	113.88

SalesDetails

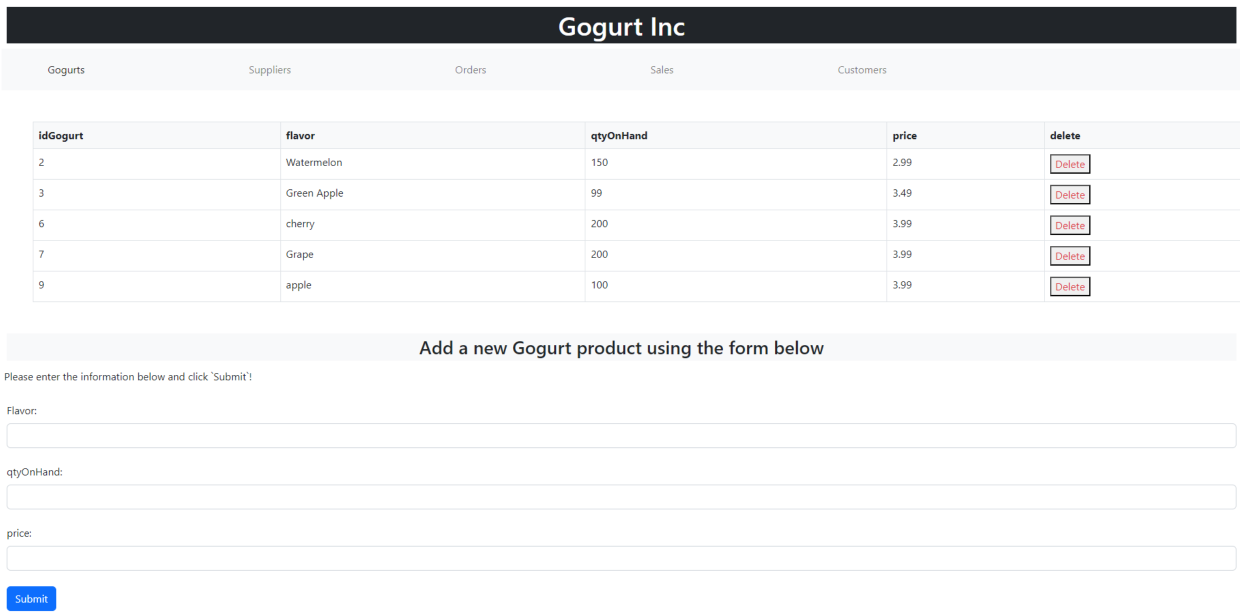
sid	gid	salesDate	orderQty	unitPrice	lineTotal
1	1	2022-02-13	4	2.99	11.96
1	2	2022-02-13	3	2.99	8.97
2	3	2022-11-13	6	3.49	20.94
3	2	2021-12-17	8	2.99	23.95

OrderDetails

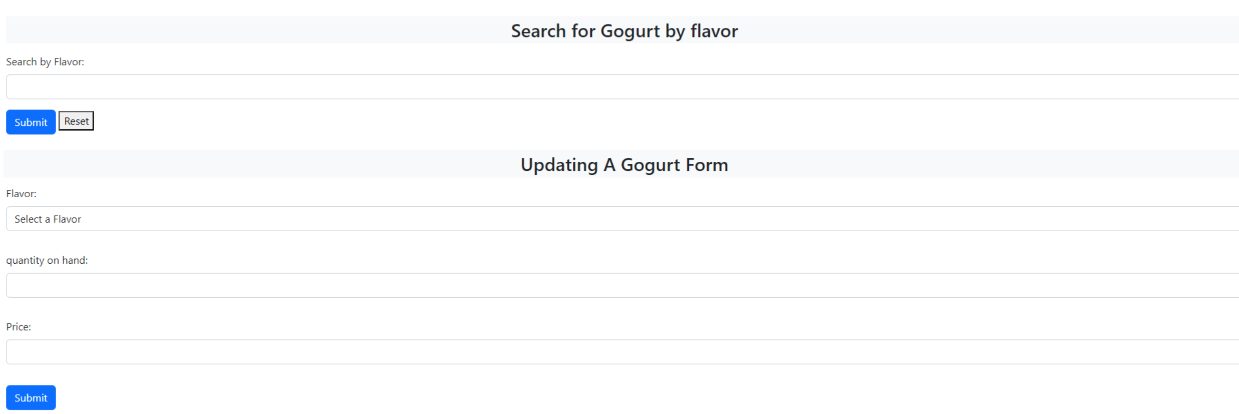
oid	gid	orderDate	orderQty	unitPrice	lineTotal
1	1	2021-12-16	16	1.99	31.84
1	2	2021-12-16	14	1.99	27.86
2	3	2022-06-17	22	2.06	45.32
2	1	2022-10-20	110	1.88	206.80

F) Screen Captures

READ/BROWSE/DISPLAY/ADD/DELETE Gogurts/Index page



UPDATE/SEARCH Gogurts/Index page



READ/BROWSE/DISPLAY/ADD/DELETE Suppliers page

Suppliers

Gogurts Suppliers Orders Sales Customers

Suppliers

idSupplier	supplierName	email	delete
1	Good Yogurt Co.	goodyogurt@sample.com	Delete
2	All Curds Inc.	allcurds@sample.com	Delete
3	South Sides Products	sideproducts@sample.com	Delete

Adding Supplier Data using an HTML form

To add a new Supplier, please enter the information below and click `Submit`!

Supplier Name: [ ] Email: [ ] Submit

READ/BROWSE/DISPLAY/ADD/DELETE Orders page

Orders

Gogurts Suppliers Orders Sales Customers

Orders

idOrder	idSupplier	orderDate	delete
1	1	Sat Mar 05 2022 00:00:00 GMT-0800 (Pacific Standard Time)	Delete
2	2	Tue Dec 14 2021 00:00:00 GMT-0800 (Pacific Standard Time)	Delete
3	3	Sat Nov 12 2022 00:00:00 GMT-0800 (Pacific Standard Time)	Delete

Orders Details

idOrderDetail	oid	gid	orderDate	orderQty	unitPrice	lineTotal
2	1	2	Thu Dec 16 2021 00:00:00 GMT-0800 (Pacific Standard Time)	14	1.99	27.86
3	2	3	Fri Jun 17 2022 00:00:00 GMT-0700 (Pacific Daylight Time)	22	2.06	45.32

Adding Data with AJAX

To add a new sale, please fill out the information below and click 'Submit'!

Enter associated supplier (supplier ID): Select a Supplier Gogurt ID: Select a Gogurt Order date: mm/dd/yyyy Order Quantity: [ ] Unit price: [ ] SubTotal: [ ] Submit

READ/BROWSE/DISPLAY/ADD/DELETE Sales page

Sales

Gogurts Suppliers Orders Sales Customers

Sales

idSale	idCustomer	shipDate	subTotal	delete
1	1	Sun Feb 20 2022 00:00:00 GMT-0800 (Pacific Standard Time)	100.25	Delete
2	1	Sat Dec 10 2022 00:00:00 GMT-0800 (Pacific Standard Time)	235.98	Delete
4	3	Fri Jan 06 2023 00:00:00 GMT-0800 (Pacific Standard Time)	113.88	Delete

Sales Details

idSalesDetail	sid	gid	salesDate	orderQty	unitPrice	lineTotal
2	1	2	Sun Feb 13 2022 00:00:00 GMT-0800 (Pacific Standard Time)	3	2.99	8.97
3	2	3	Sun Nov 13 2022 00:00:00 GMT-0800 (Pacific Standard Time)	6	3.49	20.94

Adding Data with AJAX

To add a new sale, please fill out the information below and click 'Submit'!

Customer ID: Select a Customer Gogurt ID: Select a Gogurt Ship date: mm/dd/yyyy Order Quantity: [ ] Unit price: [ ] SubTotal: [ ] Submit

READ/BROWSE/DISPLAY/ADD/DELETE Customers page

Customers

Gogurts Suppliers Orders Sales Customers

Customers

idCustomer	customerName	email	address	delete
1	Jerry West	jwest@sample.com	123 Nice Ave	Delete
3	Sara South	ssouth@sample.com	633 Yolo Dr	Delete
4	dfd	sdf@sdf	sfd	Delete

Adding Data with AJAX

To add a new customer, please enter their information below and click 'Submit'!

Customer Name: Email: Address: Submit